Department of Computer Science and Engineering

Bangladesh University of Business and Technology (BUBT)

CSE 498: Literature Review Records

Capstone Project Title	Adaptive Client Selection for Multimodal Federated Vision Foundation Models in Alzheimer's Disease Prediction
Supervisor Name & Designation	Shamim Ahmed, Assistant Professor Dept. of CSE
Course Teacher's Name & Designation	Dr. Md. Rajibul Islam & Associate Professor

Member 1

Student's Id and Name	[20234103386 & Rahmatul Munim]

Aspects	**Paper # 1**
Title / Question *(What is problem statement?)*	Title of paper: Alzheimer's Disease Detection using Deep Learning and MRI Image Analysis. Problem statement: Early and accurate diagnosis of Alzheimer's disease is difficult using traditional clinical assessment methods, and manual analysis of MRI scans by radiologists is time-consuming and prone to human error.
Objectives / Goal *(What is looking for?)*	Identify a better model to overcome the problem that will help to detect Alzheimer's disease at an early stage with higher accuracy from brain MRI images. It will help to fulfill the requirement of timely diagnosis and treatment planning in clinical practice.
Methodology/Theory *(How to find the solution?)*	Authors use a Convolutional Neural Network (CNN) based deep learning model to classify brain MRI images into different stages of Alzheimer's disease, where the model had convolutional layers, pooling layers, and fully connected dense layers. Also, they use transfer learning with a pre-trained VGG16 architecture to improve classification performance on a limited dataset.

Aspects	**Paper # 1 (cont.)**
Software Tools *(What program/software is used for design, coding and simulation?)*	Authors use TensorFlow and Keras framework in Python.
Test / Experiment *(How to test and characterize the design/prototype?)*	For classification of Alzheimer's disease stages, authors use a CNN based model combined with transfer learning. The model uses multiple convolutional layers, max-pooling layers, and dropout layers to prevent overfitting. Images were resized to 224x224 pixels, with a learning rate of 0.0001 and dropout set to 0.5 to overcome the overfitting problem.
Simulation/Test Data *(What parameters are determined?)*	Authors used the ADNI (Alzheimer's Disease Neuroimaging Initiative) dataset to obtain the experimental MRI image data. The dataset had 6400 samples total, with 5120 for training and 1280 for testing. The images were categorized into four classes: Non-Demented, Very Mild Demented, Mild Demented, and Moderate Demented.
Result / Conclusion *(What was the final result?)*	Authors use Accuracy, Precision, Recall, and F1-Score to compare performance between the proposed CNN-transfer learning model and a traditional CNN model without transfer learning, using the test dataset for the performance evaluation. Authors claim their proposed model achieved higher classification accuracy than the traditional CNN model.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Authors experimented with a class-imbalanced dataset where the Moderate Demented class had significantly fewer samples. They mention it may become more challenging to generalize the model to real-world clinical data with even greater imbalance and noise.
Terminology *(List the common basic words frequently used in this research field)*	Alzheimer's disease, MRI image, CNN, transfer learning, deep learning, accuracy, precision, recall.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	The authors proposed a model to solve the problem mentioned in the problem statement. Based on their implementation and result analysis, the proposed model performs well for multi-class classification, but may face complexity when applied to highly imbalanced or noisy real-world clinical datasets.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can try to use the authors' CNN based transfer learning proposed model for the brain disease classification case in our project.

Aspects	Paper # 2
Title / Question *(What is problem statement?)*	Title of paper: Accurate Detection of Alzheimer's Disease Using Lightweight Deep Learning Model on MRI Data. Problem statement: Alzheimer's disease (AD) is a neurodegenerative disorder, and early detection is crucial since existing deep models tend to be complex and time-consuming to run.
Objectives / Goal *(What is looking for?)*	Propose an improved lightweight deep learning model that can accurately detect Alzheimer's disease from MRI images without requiring deeper, more complex network layers.
Methodology/Theory *(How to find the solution?)*	The authors build a lightweight deep learning model consisting of only seven layers that combines feature extraction and classification into a single stage, eliminating the need for traditional separate feature-extraction and classification steps.
Software Tools *(What program/software is used for design, coding and simulation?)*	A deep learning framework was used to implement the lightweight seven-layer model (specific library/version not detailed in the available abstract).
Test / Experiment *(How to test and characterize the design/prototype?)*	The proposed lightweight model was tested on both binary classification (AD vs. non-AD) and multi-classification (multiple dementia stages) tasks, and compared with results reported by previous, larger deep learning models.
Simulation/Test Data *(What parameters are determined?)*	The model was evaluated using a publicly available Kaggle MRI dataset that is relatively small in size (about 36 megabytes) but contains a large number of image records.
Result / Conclusion *(What was the final result?)*	The proposed model achieved an overall accuracy of 99.22% for binary classification and 95.93% for multi-classification tasks, outperforming previously reported deep learning models on the same type of dataset.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The authors note that the performance of deep learning models for AD detection can be affected by the size, quality, and diversity of the available training dataset.
Terminology *(List the common basic words frequently used in this research field)*	Alzheimer's disease, MRI, lightweight deep learning, binary classification, multi-classification.

Aspects	Paper # 2 (cont.)
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Compared to heavier, deeper CNN architectures used in other AD-detection studies, this lightweight seven-layer model achieves competitive or better accuracy while being less computationally expensive, though it is validated mainly on one Kaggle dataset.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider adopting a lightweight, single-stage CNN design (combining feature extraction and classification) to reduce computational cost in our own disease-detection experiments.

Aspects	Paper # 3
Title / Question *(What is problem statement?)*	Title of paper: Leveraging Deep Learning for Early Detection of Alzheimer's Disease from MRI Scans. Problem statement: Early detection of Alzheimer's disease from MRI remains a significant diagnostic challenge, and existing models need improvement to reliably identify early AD signatures.
Objectives / Goal *(What is looking for?)*	Comprehensively review six state-of-the-art deep learning models for Alzheimer's detection and propose a new architecture based on InceptionV3 that improves prediction of AD stages from MRI data.
Methodology/Theory *(How to find the solution?)*	The authors first review six existing deep learning models, then design a new architecture on top of the InceptionV3 base model, using convolutional neural network layers to extract brain structure/function patterns and applying data augmentation to reduce class imbalance.
Software Tools *(What program/software is used for design, coding and simulation?)*	InceptionV3 pretrained CNN architecture implemented using a standard deep learning framework (specific library not detailed in the abstract).
Test / Experiment *(How to test and characterize the design/prototype?)*	The proposed and base InceptionV3 models were tested on a large-scale MRI dataset covering four stages of dementia, and their performance was measured using Area Under the Curve (AUC) before and after data augmentation.
Simulation/Test Data *(What parameters are determined?)*	A large-scale MRI dataset comprising four dementia stages was used, with data augmentation applied specifically to correct class imbalance among the stages.

Aspects	**Paper # 3 (cont.)**
Result / Conclusion *(What was the final result?)*	The base InceptionV3 model achieved 82% AUC; after data augmentation this improved to 87% AUC; the authors' proposed enhanced architecture built on InceptionV3 further improved performance to 88% AUC.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Class imbalance across the four dementia stages was identified as a key obstacle, which the authors addressed through data augmentation.
Terminology *(List the common basic words frequently used in this research field)*	Convolutional Neural Network (CNN), InceptionV3, Area Under Curve (AUC), dementia staging, data augmentation.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	The stepwise improvement from base InceptionV3 to augmented InceptionV3 to the proposed custom architecture shows that both data augmentation and architectural modification meaningfully improve early AD detection performance.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use InceptionV3-based transfer learning together with data augmentation as a strategy to address class imbalance in our own imaging-based classification task.

Aspects	**Paper # 4 (cont.)**

Aspects	**Paper # 4**
Title / Question *(What is problem statement?)*	Title of paper: Deep Learning-based Classification of MRI Images for Early Detection and Staging of Alzheimer's Disease. Problem statement: There is currently no cure for Alzheimer's disease, so early identification is crucial to allow treatment before significant brain damage occurs.
Objectives / Goal *(What is looking for?)*	Study various deep learning techniques to address the challenge of early Alzheimer's disease detection and staging using structural MRI (sMRI) images as biomarkers.
Methodology/Theory *(How to find the solution?)*	The authors use the GoogLeNet architecture to extract features from each MRI scan; these extracted features are then passed into a feed-forward neural network (FFNN) which predicts the Alzheimer's disease stage.
Software Tools *(What program/software is used for design, coding and simulation?)*	GoogLeNet (for feature extraction) combined with a feed-forward neural network (FFNN) for classification.
Test / Experiment *(How to test and characterize the design/prototype?)*	The FFNN model, using GoogLeNet-derived features, was trained over multiple epochs with a small batch size to ensure robust performance on unseen data, and evaluated using accuracy, sensitivity, precision, and specificity.
Simulation/Test Data *(What parameters are determined?)*	A balanced MRI image dataset of 12,936 images was used, which the authors note is larger than datasets used in many previous studies, to better capture the subtle similarities between early AD stages.
Result / Conclusion *(What was the final result?)*	The proposed GoogLeNet + FFNN model achieved 98.37% accuracy, 98.39% sensitivity, 98.50% precision, and 99.45% specificity.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The authors mention that the similarity in MRI characteristics between early AD stages required a larger and more carefully balanced dataset than used in earlier studies.
Terminology *(List the common basic words frequently used in this research field)*	Structural MRI (sMRI), GoogLeNet, feed-forward neural network (FFNN), AD staging.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Using a pretrained CNN purely as a feature extractor and pairing it with a separate FFNN classifier is shown to be an effective and accurate hybrid strategy for multi-stage AD classification.

Aspects	**Paper # 4 (cont.)**
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider a similar hybrid design, using a pretrained CNN as a feature extractor feeding into a simpler classifier, for staged disease classification in our project.

Aspects	**Paper # 5**
Title / Question *(What is problem statement?)*	Title of paper: Heart Disease Prediction Using Machine Learning Techniques. Problem statement: Cardiovascular diseases are responsible for a majority of deaths globally, making accurate prediction models important for prevention.
Objectives / Goal *(What is looking for?)*	Perform comparative research on the accuracy of several machine learning algorithms to determine which model best predicts heart disease, to be developed further into a best-accuracy prediction system.
Methodology/Theory *(How to find the solution?)*	The authors compare five supervised machine learning algorithms: Random Forest, Support Vector Machine, AdaBoost Classifier, Logistic Regression, and Decision Tree Classifier, evaluating each with a training/testing split.
Software Tools *(What program/software is used for design, coding and simulation?)*	A Python-based machine learning workflow was used to implement and compare the classifiers (specific libraries not detailed in the available excerpt).
Test / Experiment *(How to test and characterize the design/prototype?)*	Each of the five algorithms was trained and tested on the same heart disease dataset, and their prediction accuracies were compared to identify the strongest performing model.
Simulation/Test Data *(What parameters are determined?)*	The dataset used contained records of heart disease and normal (healthy) cases with no missing values, allowing direct use of supervised learning without extensive imputation.
Result / Conclusion *(What was the final result?)*	Among the five algorithms compared, Random Forest achieved the best accuracy at 85.22%, outperforming SVM, AdaBoost, Logistic Regression, and Decision Tree on this dataset.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The authors note the work is intended to be developed further to reach even higher prediction accuracy, implying that further feature engineering or tuning is still needed.

Aspects	**Paper # 5 (cont.)**
Terminology *(List the common basic words frequently used in this research field)*	Heart disease prediction, Random Forest, Support Vector Machine, AdaBoost, Decision Tree.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Among classical machine learning approaches tested on a relatively small, clean heart disease dataset, ensemble-based Random Forest clearly outperformed the other individual classifiers.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use Random Forest as a strong baseline classical machine learning model when comparing against deep learning approaches in our own disease-prediction work.

Member 2

Student's Id and Name	[20234103384 & Mehedi Hassan]

Aspects	**Paper # 1**
Title / Question *(What is problem statement?)*	Title of paper: Using Machine Learning to Predict Heart Disease. Problem statement: Heart disease accounts for a large share of global deaths (about 31%), so early and accurate prediction is necessary to reduce mortality.
Objectives / Goal *(What is looking for?)*	Predict the likelihood of a person developing heart disease using clinical features such as cholesterol, blood pressure, age, and sex, and identify which machine learning algorithm gives the highest accuracy.
Methodology/Theory *(How to find the solution?)*	The authors implement and compare six machine learning algorithms – Logistic Regression, Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Random Forest, and XGBoost – and also test the effect of combining two separate datasets into one larger dataset.
Software Tools *(What program/software is used for design, coding and simulation?)*	Python-based machine learning stack used for data preprocessing, feature engineering, and model training (specific library names not detailed in the excerpt).
Test / Experiment *(How to test and characterize the design/prototype?)*	Two datasets were tested separately, then combined; for each configuration, all six algorithms were trained and their accuracies compared, with data preprocessing and feature engineering applied to improve results.
Simulation/Test Data *(What parameters are determined?)*	The first dataset was the UCI heart disease dataset (303 records, 14 attributes). The second was a Kaggle dataset combining five popular heart-disease datasets (1190 records, 11 features). The combined dataset totalled 1493 records with 12 attributes.
Result / Conclusion *(What was the final result?)*	On the UCI dataset, SVM achieved the highest accuracy at 92%. On the Kaggle dataset, Random Forest achieved 94.12% accuracy. On the combined dataset, Random Forest achieved 93.31% accuracy.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Combining two datasets with different feature sets required careful preprocessing and feature alignment before the models could be trained on the merged data.

Aspects	Paper # 1 (cont.)
Terminology *(List the common basic words frequently used in this research field)*	UCI dataset, Kaggle dataset, feature engineering, ensemble learning, XGBoost.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This study shows that model performance varies noticeably by dataset, and that combining multiple datasets does not always outperform the best single-dataset result, though Random Forest performed consistently well across configurations.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use a similar strategy of testing multiple public datasets separately and combined, to check whether merging improves or reduces performance for our chosen disease-prediction task.

Aspects	Paper # 2
Title / Question *(What is problem statement?)*	Title of paper: Prediction of Heart Disease Using Machine Learning Algorithms. Problem statement: Determining how likely a person is to develop heart disease is an important but technically difficult task in the medical field due to the many interacting risk factors involved.
Objectives / Goal *(What is looking for?)*	Investigate whether an ensemble voting classifier can outperform individual machine learning algorithms in predicting the presence of heart disease.
Methodology/Theory *(How to find the solution?)*	The authors implement KNN, Logistic Regression, Naive Bayes, and SVM as individual classifiers, and additionally combine AdaBoost and XGBoost into an ensemble voting classifier for comparison.
Software Tools *(What program/software is used for design, coding and simulation?)*	Python-based machine learning implementation of the individual classifiers and the ensemble voting classifier (specific libraries not detailed in the excerpt).
Test / Experiment *(How to test and characterize the design/prototype?)*	All individual classifiers and the voting ensemble were trained and evaluated on the same dataset, and their prediction accuracies were compared to determine the best-performing approach.
Simulation/Test Data *(What parameters are determined?)*	A standard heart disease dataset with patient clinical attributes was used for training and testing (exact size not detailed in the available excerpt).
Result / Conclusion *(What was the final result?)*	The ensemble voting classifier achieved the highest prediction accuracy of 98.3%, outperforming all of the individual machine learning methods tested.

Aspects	Paper # 2 (cont.)
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The paper does not detail specific obstacles beyond the general difficulty of heart disease prediction due to its dependence on many interacting clinical factors.
Terminology *(List the common basic words frequently used in this research field)*	Voting classifier, ensemble learning, AdaBoost, XGBoost, KNN.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This study supports the trend seen across the reviewed heart-disease papers: ensemble or voting-based approaches tend to outperform single classical classifiers on tabular clinical data.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider building an ensemble/voting classifier that combines multiple base models to potentially improve prediction accuracy in our own project.

Aspects	Paper # 3
Title / Question *(What is problem statement?)*	Title of paper: Fundus Image-Based Eye Disease Detection Using EfficientNetB3 Architecture. Problem statement: Retinal fundus images vary in noise, illumination, resolution and quality, making automated multi-class eye disease diagnosis a challenging task for traditional classifiers.
Objectives / Goal *(What is looking for?)*	Develop a deep learning classification model that can automatically categorize fundus images into cataract, diabetic retinopathy, glaucoma, and healthy classes to support automated retinal disease diagnosis.
Methodology/Theory *(How to find the solution?)*	The authors fine-tune a pretrained EfficientNetB3 architecture using transfer learning, apply data augmentation to handle image quality variation, and optimize training using the Adam optimizer with a cosine annealing learning-rate scheduler.
Software Tools *(What program/software is used for design, coding and simulation?)*	EfficientNetB3 pretrained architecture implemented with a deep learning framework, using the Adam optimizer and a cosine annealing scheduler.
Test / Experiment *(How to test and characterize the design/prototype?)*	The fine-tuned EfficientNetB3 model was tested on a held-out portion of the retinal image dataset and evaluated across multiple metrics including accuracy, precision, recall, F1-score, Dice score, Jaccard index, and Matthews Correlation Coefficient (MCC).

Aspects	Paper # 3 (cont.)
Simulation/Test Data *(What parameters are determined?)*	A publicly available Kaggle retinal fundus image dataset was used, categorized into four classes: cataract, diabetic retinopathy, glaucoma, and healthy.
Result / Conclusion *(What was the final result?)*	The proposed model achieved a classification accuracy of 95.12%, with precision of 95.21%, recall of 94.88%, F1-score of 95.00%, Dice score of 94.91%, Jaccard index of 91.2%, and MCC of 0.925.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Variability in retinal image noise, illumination, and resolution across patients and equipment, along with intra-class diversity and inter-class similarity between certain eye disorders, were noted as key challenges.
Terminology *(List the common basic words frequently used in this research field)*	Fundus imaging, EfficientNetB3, transfer learning, cataract, glaucoma, diabetic retinopathy.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This paper demonstrates that a moderately-sized, efficient pretrained architecture (EfficientNetB3) combined with careful augmentation and optimizer scheduling can achieve strong multi-class accuracy on noisy real-world retinal images.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider using an EfficientNet-family backbone with cosine annealing scheduling as a computationally efficient option for our own image classification component.

Aspects	Paper # 4 (cont.)

Aspects	Paper # 4
Title / Question *(What is problem statement?)*	Title of paper: An EfficientNet-B5-Based Deep Learning Model for Multilabel Classification of Fundus Images. Problem statement: Ophthalmologists rely heavily on manual fundus image examination to diagnose retinal diseases; early detection is important to prevent blindness but manual screening does not scale well.
Objectives / Goal *(What is looking for?)*	Develop an automated diagnostic approach using deep learning to identify diseases in fundus images with a high prediction rate, focusing on classifying diabetic retinopathy severity levels.
Methodology/Theory *(How to find the solution?)*	The authors build a model based on the EfficientNet-B5 architecture, apply image preprocessing (conversion to RGB, resizing to 224x224, and Gaussian blur filtering), and use 10-fold cross-validation to train and validate the model.
Software Tools *(What program/software is used for design, coding and simulation?)*	EfficientNet-B5 architecture implemented with standard image preprocessing utilities (RGB conversion, resizing, Gaussian blur filtering).
Test / Experiment *(How to test and characterize the design/prototype?)*	The model was trained and validated using 10-fold cross-validation on the preprocessed fundus images, classifying each image as normal, non-proliferative diabetic retinopathy (NPDR), or proliferative diabetic retinopathy (PDR).
Simulation/Test Data *(What parameters are determined?)*	The DDR (Diabetic Retinopathy) dataset was used, with a specific focus on three classes: normal, NPDR, and PDR.
Result / Conclusion *(What was the final result?)*	The enhanced EfficientNet-B5 model achieved a validation accuracy of 96.04% and a training accuracy of 99.54%.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The paper implies that distinguishing between the different severity levels of diabetic retinopathy (normal, NPDR, PDR) is a subtle classification challenge requiring careful preprocessing and cross-validation to avoid overfitting.
Terminology *(List the common basic words frequently used in this research field)*	NPDR, PDR, EfficientNet-B5, cross-validation, Gaussian blur filtering.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	The use of systematic preprocessing (resizing, blur filtering) together with k-fold cross-validation helped the EfficientNet-B5 model achieve high and reliable accuracy for diabetic retinopathy severity staging.

Aspects	Paper # 4 (cont.)
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can adopt a similar preprocessing pipeline (standard resizing, noise filtering) combined with k-fold cross-validation to make our own model's evaluation more robust.

Aspects	Paper # 5
Title / Question *(What is problem statement?)*	Title of paper: Classification of Fundus Images Based on Deep Learning for Detecting Eye Diseases. Problem statement: Most prior studies examined only one or two eye diseases at a time compared against healthy images; simultaneous detection of multiple major eye diseases from fundus photographs was not well addressed.
Objectives / Goal *(What is looking for?)*	Achieve multi-categorical classification of three major eye diseases – diabetic retinopathy (DR), glaucoma (GLC), and age-related macular degeneration (AMD) – together with healthy cases, from a single fundus image dataset.
Methodology/Theory *(How to find the solution?)*	The authors use an optimal residual deep neural network (ResNet-based) combined with image preprocessing techniques such as shrinking the region of interest and iso-luminance plane contrast-limited adaptive histogram equalization, plus data augmentation.
Software Tools *(What program/software is used for design, coding and simulation?)*	A ResNet-based deep neural network implemented with region-of-interest preprocessing and contrast-limited histogram equalization techniques.
Test / Experiment *(How to test and characterize the design/prototype?)*	The residual network was tested on multiple currently available public fundus image datasets, and performance was measured through per-class accuracy and specificity across the four categories (healthy, DR, GLC, AMD).
Simulation/Test Data *(What parameters are determined?)*	Currently available public fundus image datasets covering healthy eyes and the three target eye diseases (DR, GLC, AMD) were combined for training and evaluation.

Aspects	Paper # 5 (cont.)
Result / Conclusion *(What was the final result?)*	The model achieved a peak accuracy of 91.16% and an average accuracy of 85.79% across categories; specificity values were 90.06% for healthy, 99.63% for glaucoma, 99.82% for AMD, and 91.90% for diabetic retinopathy.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Simultaneously distinguishing multiple eye diseases (rather than one disease vs. healthy) is inherently more difficult, since features of different diseases can overlap in fundus images.
Terminology *(List the common basic words frequently used in this research field)*	ResNet, multi-categorical classification, specificity, age-related macular degeneration (AMD).
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Compared to studies focusing on a single disease, this multi-categorical approach achieves reasonably strong specificity per disease class, though overall accuracy is somewhat lower than simpler binary classification studies – reflecting the added difficulty of multi-disease detection.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider a similar residual-network-based multi-disease classification setup if our project needs to detect more than one condition from the same type of image.

Member 3

Student's Id and Name	[20234103379 & Mostafa Monoar]

Aspects	Paper # 1
Title / Question *(What is problem statement?)*	Title of paper: Brain tumor classification using MRI images and deep learning techniques. Problem statement: Prior brain tumor detection studies mostly focused on binary classification (tumor vs. no tumor); a more comprehensive multiclass solution distinguishing tumor types was needed.
Objectives / Goal *(What is looking for?)*	Develop a model that can accurately detect and classify different types of brain tumors – glioma, meningioma, pituitary tumors – as well as normal brain scans, moving beyond simple binary classification.
Methodology/Theory *(How to find the solution?)*	The authors use a convolutional neural network (CNN) architecture built on a pretrained VGG16 base model, combine multiple diverse public MRI datasets for broader representation, and apply data augmentation to expand the training set.
Software Tools *(What program/software is used for design, coding and simulation?)*	Pretrained VGG16 CNN architecture with additional custom layers and fine-tuning.
Test / Experiment *(How to test and characterize the design/prototype?)*	The fine-tuned VGG16-based CNN model was tested on a combined, augmented dataset covering all four classes (glioma, meningioma, pituitary tumor, normal) and compared against results from other similar published works.
Simulation/Test Data *(What parameters are determined?)*	Diverse public brain MRI datasets were combined and augmented to produce a total of 17,136 MRI images across the four classes.
Result / Conclusion *(What was the final result?)*	The proposed model achieved an accuracy of 99.24%, higher than other comparable works reviewed by the authors.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Dataset limitations (size and diversity) were addressed by combining multiple public datasets and applying data augmentation, which the authors identify as key contributors to their improved performance.
Terminology *(List the common basic words frequently used in this research field)*	Glioma, meningioma, pituitary tumor, VGG16, multiclass classification.

Aspects	Paper # 1 (cont.)
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Combining several public datasets with augmentation and fine-tuning a pretrained VGG16 model proved highly effective for multiclass brain tumor classification, outperforming narrower binary-classification approaches.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider combining multiple public MRI datasets and applying augmentation to increase training data diversity in our own brain-related classification task.

Aspects	Paper # 2
Title / Question *(What is problem statement?)*	Title of paper: Brain Tumor Detection and Classification Using a Hyperparameter Tuned Convolutional Neural Network. Problem statement: Manual MRI-based brain tumor diagnosis requires significant medical professional involvement and is time-consuming; an automated, accurate early detection system is needed.
Objectives / Goal *(What is looking for?)*	Develop an advanced predictive CNN-based model that accurately classifies brain tumors (benign or malignant, and by tumor type) using MRI scans, with minimal need for medical expert input.
Methodology/Theory *(How to find the solution?)*	The authors propose a novel CNN model and apply systematic hyperparameter tuning to automate tumor detection and improve classification accuracy across four tumor-related classes.
Software Tools *(What program/software is used for design, coding and simulation?)*	A convolutional neural network (CNN) implemented with a hyperparameter tuning process (specific tuning library not detailed in the available excerpt).
Test / Experiment *(How to test and characterize the design/prototype?)*	The hyperparameter-tuned CNN was trained and tested on the brain tumor dataset to classify images into glioma, meningioma, pituitary tumor, and no-tumor classes, comparing different hyperparameter configurations.
Simulation/Test Data *(What parameters are determined?)*	A dataset of around 7000 brain MRI images was used, classified into four labels: glioma, meningioma, pituitary, and no tumor.

Aspects	**Paper # 2 (cont.)**
Result / Conclusion *(What was the final result?)*	The hyperparameter-tuned CNN model showed improved classification performance over untuned baseline CNN configurations for distinguishing the four tumor-related classes (exact accuracy figures are reported in the full paper).
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Selecting an optimal set of hyperparameters without causing overfitting was identified as a central challenge addressed through the tuning process.
Terminology *(List the common basic words frequently used in this research field)*	Hyperparameter tuning, CNN, benign/malignant classification, tumor grading.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This study emphasizes that systematic hyperparameter tuning, rather than just architecture choice, is an important factor for improving CNN-based tumor classification accuracy.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can apply a systematic hyperparameter tuning process (learning rate, number of layers, batch size) when building our own CNN model, rather than relying on default settings.

Aspects	Paper # 3 (cont.)

Aspects	Paper # 3
Title / Question *(What is problem statement?)*	Title of paper: Accurate Brain Tumor Detection and Classification System Based on MRI Images Using Deep Learning Technology. Problem statement: Non-invasive and accurate diagnosis using MRI is crucial for effective brain tumor treatment planning, but achieving high accuracy requires careful CNN design choices.
Objectives / Goal *(What is looking for?)*	Develop a CNN-based system for the detection and classification of brain tumors versus normal cases based on a public Kaggle MRI dataset, exploring various CNN design options to maximize accuracy.
Methodology/Theory *(How to find the solution?)*	The authors explore multiple CNN design options including different optimizers (Adam, AdaDelta, and SGD), layer configurations, receptive field sizes, stride, kernel size, and padding settings to identify the best-performing architecture.
Software Tools *(What program/software is used for design, coding and simulation?)*	Convolutional Neural Network (CNN) implemented and tested with multiple optimizers – Adam, AdaDelta, and SGD.
Test / Experiment *(How to test and characterize the design/prototype?)*	The CNN system was tested with different combinations of optimizers and architectural settings on the Kaggle brain tumor dataset, comparing testing accuracy across configurations.
Simulation/Test Data *(What parameters are determined?)*	The Kaggle dataset 'Brain Tumor Classification MRI' was used, consisting of 3264 MRI images: 2764 brain tumor cases across three tumor types, and 500 normal (no-tumor) images.
Result / Conclusion *(What was the final result?)*	The proposed CNN system achieved a testing accuracy of 100%, demonstrating very high efficiency in recognizing brain tumors on this dataset.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The dataset used has significantly more tumor images (2764) than normal images (500), which represents a class-imbalance concern that could affect generalization to new data.
Terminology *(List the common basic words frequently used in this research field)*	CNN, optimizer tuning (Adam, AdaDelta, SGD), receptive field, Kaggle dataset.

Aspects	Paper # 3 (cont.)
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Through an exhaustive search of optimizer and architecture settings, this study achieved a very high reported accuracy, though the class imbalance in the dataset used raises questions about generalization compared to more balanced studies.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can systematically compare multiple optimizers and architecture settings, as done here, while also being careful to check for class imbalance in our own dataset.

Aspects	Paper # 4
Title / Question *(What is problem statement?)*	Title of paper: Using Deep Learning for Image-Based Plant Disease Detection. Problem statement: Crop diseases remain a major threat to global food supply, and manual disease diagnosis does not scale to meet the needs of farmers worldwide.
Objectives / Goal *(What is looking for?)*	Demonstrate the technical feasibility of a deep learning approach for automatic plant disease diagnosis through image recognition, jointly classifying crop species and disease status.
Methodology/Theory *(How to find the solution?)*	The authors train a deep convolutional neural network on a large public dataset of leaf images to jointly classify both the crop species and the disease (or healthy) status of each leaf image.
Software Tools *(What program/software is used for design, coding and simulation?)*	A deep convolutional neural network implemented using a standard deep learning framework (specific library not detailed in the abstract).
Test / Experiment *(How to test and characterize the design/prototype?)*	The trained CNN was tested on held-out images from the PlantVillage dataset across 38 crop-disease classes to measure classification accuracy.
Simulation/Test Data *(What parameters are determined?)*	The PlantVillage public dataset was used, comprising 54,306 images of diseased and healthy plant leaves across 38 classes, covering 14 crop species and 26 diseases.
Result / Conclusion *(What was the final result?)*	The deep convolutional neural network achieved an accuracy of over 99% in classifying crop species and disease status.

Aspects	**Paper # 4 (cont.)**
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	A commonly noted limitation of PlantVillage-based studies (including this one) is that the images were captured under controlled/lab conditions, which may not fully represent real-world field conditions.
Terminology *(List the common basic words frequently used in this research field)*	PlantVillage dataset, convolutional neural network (CNN), crop-disease pair, image-based diagnosis.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	As one of the earliest large-scale demonstrations of deep learning for plant disease diagnosis, this paper set an important accuracy benchmark (over 99%) that many later leaf-disease studies compare against.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use the PlantVillage dataset and this paper's reported accuracy as a baseline benchmark for our own leaf disease detection work.

Member 4

Student's Id and Name	[20234103371 & Mubashsher Salek]

Aspects	Paper # 1
Title / Question *(What is problem statement?)*	Title of paper: Deep Learning-Based Leaf Disease Detection in Crops Using Images for Agricultural Applications. Problem statement: Plant disease can cause significant losses in food production, so accurate and efficient identification techniques are needed to minimize economic losses.
Objectives / Goal *(What is looking for?)*	Achieve efficient plant disease identification by fine-tuning the hyperparameters of several popular pretrained CNN models and comparing their classification performance.
Methodology/Theory *(How to find the solution?)*	The authors fine-tune the hyperparameters of four pretrained CNN models – DenseNet-121, ResNet-50, VGG-16, and Inception V4 – for the plant disease classification task, then compare their performance.
Software Tools *(What program/software is used for design, coding and simulation?)*	Pretrained CNN architectures: DenseNet-121, ResNet-50, VGG-16, and Inception V4, with hyperparameter fine-tuning.
Test / Experiment *(How to test and characterize the design/prototype?)*	All four pretrained models were fine-tuned and evaluated on the same dataset, using classification accuracy, sensitivity, specificity, and F1-score, and compared against other state-of-the-art studies.
Simulation/Test Data *(What parameters are determined?)*	The popular PlantVillage dataset was used, containing 54,305 image samples across 38 classes of different plant disease species.
Result / Conclusion *(What was the final result?)*	DenseNet-121 achieved the highest classification accuracy of 99.81%, outperforming ResNet-50, VGG-16, Inception V4, and other state-of-the-art models compared in the study.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Fine-tuning and comparing four different pretrained architectures required considerable computational resources and careful hyperparameter search for each model.
Terminology *(List the common basic words frequently used in this research field)*	DenseNet-121, ResNet-50, VGG-16, Inception V4, hyperparameter fine-tuning, transfer learning.

Aspects	Paper # 1 (cont.)
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Compared with the earlier PlantVillage benchmark study, this work shows that careful hyperparameter fine-tuning of a modern architecture (DenseNet-121) can push accuracy even higher (99.81% vs. 99%).
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider DenseNet-121 as a strong candidate backbone architecture when comparing pretrained models for our own leaf disease classification component.

Aspects	Paper # 2
Title / Question *(What is problem statement?)*	Title of paper: Plants Leaf Diseases Detection Using Deep Learning. Problem statement: Agricultural productivity depends heavily on timely disease detection; accurate prediction helps treat leaves early and control economic loss from crop disease.
Objectives / Goal *(What is looking for?)*	Use image processing techniques together with a CNN to classify and detect plant leaf diseases accurately, while addressing the risk of model overfitting.
Methodology/Theory *(How to find the solution?)*	The authors build a CNN model and apply data augmentation, dropout, and weight regularization techniques to prevent overfitting, using the Adam optimizer; they further improve the baseline model by adding early stopping.
Software Tools *(What program/software is used for design, coding and simulation?)*	CNN implemented with the Adam optimizer, using dropout and weight regularization for overfitting prevention.
Test / Experiment *(How to test and characterize the design/prototype?)*	The CNN was tested on the PlantVillage dataset in two configurations: a baseline model, and an improved model using early stopping, comparing training and testing accuracy between the two.
Simulation/Test Data *(What parameters are determined?)*	The publicly available PlantVillage dataset was used, consisting of 15 classes: 12 disease classes and 3 healthy leaf classes.
Result / Conclusion *(What was the final result?)*	The baseline model achieved 98.08% testing accuracy and 99.24% training accuracy. After adding early stopping, accuracy improved to 98.34% on testing and 99.64% on training.

Aspects	**Paper # 2 (cont.)**
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Overfitting was the main challenge identified, which the authors addressed through dropout, weight regularization, and early stopping.
Terminology *(List the common basic words frequently used in this research field)*	Dropout, weight regularization, early stopping, PlantVillage, Adam optimizer.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This study shows that systematic overfitting-prevention techniques (dropout, regularization, early stopping) can produce small but consistent accuracy improvements over a baseline CNN, complementing architecture-focused studies like DenseNet-121 fine-tuning.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can apply a combination of dropout, weight regularization, and early stopping in our own CNN training pipeline to reduce overfitting.

Aspects	**Paper # 3**
Title / Question *(What is problem statement?)*	Title of paper: Traffic Sign Recognition and Classification Using Machine Learning and Deep Learning. Problem statement: Traffic sign recognition and classification are essential for intelligent transportation systems and autonomous vehicles, and a real-time, robust classification system is still needed.
Objectives / Goal *(What is looking for?)*	Create a real-time, robust traffic sign classification system using machine learning and deep learning techniques, benchmarked on a standard traffic sign dataset.
Methodology/Theory *(How to find the solution?)*	The authors implement CNN-based deep learning models, referencing architectures such as Inception V3, Inception-ResNet, MobileNet, ResNet50, and Xception, and apply data augmentation and dropout techniques to reduce overfitting during training.
Software Tools *(What program/software is used for design, coding and simulation?)*	CNN architectures from the Inception, ResNet, MobileNet, and Xception families implemented in a deep learning framework.

Aspects	**Paper # 3 (cont.)**
Test / Experiment *(How to test and characterize the design/prototype?)*	Models were trained and tested on the GTSRB benchmark dataset, with accuracy, precision, recall, and F1-score used as evaluation metrics, and results compared against previously reported approaches on the same dataset.
Simulation/Test Data *(What parameters are determined?)*	The German Traffic Sign Recognition Benchmark (GTSRB) dataset was used, consisting of 43 traffic sign classes with 39,209 images used for training.
Result / Conclusion *(What was the final result?)*	The study reports that CNN-based deep learning methods can achieve accuracy of up to 99.99% on the GTSRB dataset, with combined classification-and-detection approaches reaching around 99.3% accuracy in the literature reviewed.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Overfitting was identified as a key challenge, addressed through data augmentation and dropout techniques during training.
Terminology *(List the common basic words frequently used in this research field)*	GTSRB, CNN, data augmentation, dropout, real-time recognition.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This paper confirms that modern CNN-based deep learning methods substantially outperform traditional computer-vision-based approaches for traffic sign recognition, achieving near-perfect accuracy on the GTSRB benchmark.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use the GTSRB benchmark dataset along with augmentation/dropout strategies if our project includes a traffic sign or object recognition component.

Aspects	**Paper # 4**
Title / Question *(What is problem statement?)*	Title of paper: Traffic Flow Prediction Based on Deep Learning in Internet of Vehicles. Problem statement: Traditional traffic flow prediction methods suffer from overfitting and require manual intervention, and cannot support large-scale, high-dimensional urban road network data.
Objectives / Goal *(What is looking for?)*	Propose a deep-learning-based traffic flow prediction framework for large urban road networks in an Internet of Vehicles (IoV) setting, capable of handling large-scale, high-dimensional traffic data.

Aspects	**Paper # 4 (cont.)**
Methodology/Theory *(How to find the solution?)*	The authors apply feature engineering to extract features from a large traffic dataset while removing anomaly nodes, use a spectral clustering compression scheme to reduce the size of the big traffic dataset, and design a hybrid prediction model combining LSTM (Long Short-Term Memory) with a Sparse Auto-Encoder (SAE).
Software Tools *(What program/software is used for design, coding and simulation?)*	LSTM and Sparse Auto-Encoder (SAE) implemented together with a spectral clustering compression step.
Test / Experiment *(How to test and characterize the design/prototype?)*	The hybrid LSTM+SAE model was tested on large-scale urban road network traffic data and its prediction accuracy was compared with other traffic flow prediction models.
Simulation/Test Data *(What parameters are determined?)*	A large-scale urban road network traffic dataset was used within the Internet of Vehicles (IoV) context (specific public source not detailed in the available abstract).
Result / Conclusion *(What was the final result?)*	The proposed hybrid LSTM+SAE model achieved an average prediction accuracy of approximately 97.7%, outperforming the other models compared in the study.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Handling large-scale, high-dimensional urban traffic data without overfitting or requiring manual intervention was the central challenge the framework aimed to solve.
Terminology *(List the common basic words frequently used in this research field)*	Internet of Vehicles (IoV), LSTM, Sparse Auto-Encoder (SAE), spectral clustering, feature engineering.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	By combining dimensionality reduction (spectral clustering) with a hybrid LSTM+SAE model, this study effectively addresses the scalability problems that limit traditional traffic-flow prediction methods on large datasets.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider combining a dimensionality-reduction step with a hybrid deep learning model (e.g. LSTM+autoencoder) if our project needs to handle large-scale traffic or sensor data.

Member 5

Student's Id and Name	[20234103363 & Md. Jewel Rana]

Aspects	**Paper # 1**
Title / Question *(What is problem statement?)*	Title of paper: Traffic Flow Prediction Using Deep Learning Techniques (DAE, DBN, RF, LSTM Comparison). Problem statement: Urban traffic congestion is a growing problem, especially as more people commute in cities, motivating the need for accurate traffic flow prediction techniques.
Objectives / Goal *(What is looking for?)*	Predict traffic flow outcomes and compare the effectiveness of four different machine learning and deep learning techniques for this task.
Methodology/Theory *(How to find the solution?)*	The authors implement and compare four techniques for traffic flow prediction: Deep AutoEncoder (DAE), Deep Belief Network (DBN), Random Forest (RF), and Long Short-Term Memory (LSTM), all implemented in Python.
Software Tools *(What program/software is used for design, coding and simulation?)*	Python programming language used to implement Deep AutoEncoder, Deep Belief Network, Random Forest, and LSTM models.
Test / Experiment *(How to test and characterize the design/prototype?)*	Each of the four techniques was trained and tested on the same traffic dataset, with accuracy and error values compared across all four approaches to determine the best-performing method.
Simulation/Test Data *(What parameters are determined?)*	A traffic flow dataset was used to train and evaluate the four compared techniques (specific public source not detailed in the available excerpt).
Result / Conclusion *(What was the final result?)*	The LSTM-based model produced the highest accuracy of 94.3% with the lowest error value among the four techniques compared (DAE, DBN, RF, and LSTM).
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Fairly comparing four structurally different techniques (two deep learning models, one ensemble tree model, and one recurrent model) on the same dataset was a methodological challenge addressed by the authors.
Terminology *(List the common basic words frequently used in this research field)*	Deep AutoEncoder (DAE), Deep Belief Network (DBN), Random Forest (RF), LSTM.

Aspects	Paper # 1 (cont.)
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Consistent with other traffic-flow prediction papers reviewed, LSTM's ability to model sequential/time-series patterns gave it an advantage over non-sequential techniques like Random Forest for this comparison.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	Based on this and the IoV-based study, we can prioritize LSTM as our primary model choice if our project involves any traffic or time-series flow prediction component.

Aspects	Paper # 2
Title / Question *(What is problem statement?)*	Title of paper: Chatbot Recommender Systems in Tourism: A Systematic Review and a Benefit-Cost Analysis. Problem statement: There is a need to understand the practical pros and cons of using AI-based customer-service chatbots in the travel, tourism, and hospitality industry.
Objectives / Goal *(What is looking for?)*	Systematically review and synthesize existing literature on AI-based conversational chatbot systems used for customer service in tourism and hospitality, and analyze their benefits and costs.
Methodology/Theory *(How to find the solution?)*	The authors apply a systematic literature review methodology, using rigorous criteria to search, screen, extract, and synthesize articles on chatbots and tourism/hospitality, then outline the pros and cons found across the reviewed studies.
Software Tools *(What program/software is used for design, coding and simulation?)*	Systematic review methodology (search, screening, extraction, and synthesis of published articles); not a code-based implementation.
Test / Experiment *(How to test and characterize the design/prototype?)*	The review process itself served as the 'test', involving screening of relevant articles against defined inclusion criteria and synthesizing common findings across the most-cited works on chatbots in tourism.
Simulation/Test Data *(What parameters are determined?)*	The dataset consisted of a corpus of previously published, most-cited academic articles on the use of chatbots in tourism and hospitality, rather than a numerical or image dataset.

Aspects	Paper # 2 (cont.)
Result / Conclusion *(What was the final result?)*	The review synthesizes the pros and cons of AI chatbot adoption in tourism customer service, concluding that there is still considerable scope for tourism businesses to further improve their online customer service using these technologies.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Synthesizing findings across a heterogeneous set of studies, each using different methods, populations, and evaluation criteria, was a key challenge inherent to the systematic review approach.
Terminology *(List the common basic words frequently used in this research field)*	Chatbot, conversational AI, systematic review, benefit-cost analysis, customer service.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Unlike the technical, model-focused papers reviewed elsewhere in this table, this paper offers a broader, business-oriented perspective on AI chatbot adoption, which is useful for understanding practical deployment considerations rather than raw model accuracy.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use this paper's benefit-cost framing as a reference when presenting the practical value and limitations of our own AI-based system to non-technical stakeholders.

Aspects	Paper # 3
Title / Question *(What is problem statement?)*	Title of paper: The Educational Recommendation System with Artificial Intelligence Chatbot: A Case Study in Thailand. Problem statement: Students need accessible, accurate, and timely educational and program recommendations, but traditional advising methods do not scale well or provide 24/7 access.
Objectives / Goal *(What is looking for?)*	Develop a mobile AI-chatbot-based educational recommendation system to provide information and recommendations about an engineering degree program at a college in Thailand, accessible via smartphones and social media platforms.
Methodology/Theory *(How to find the solution?)*	The authors design and implement a recommendation system compatible with both iOS and Android, integrated with popular social media applications, capable of analyzing user questions and returning precise, relevant answers through an AI chatbot interface.

Aspects	Paper # 3 (cont.)
Software Tools *(What program/software is used for design, coding and simulation?)*	Mobile application development platform integrated with an AI chatbot engine (specific NLP/chatbot framework not detailed in the available excerpt).
Test / Experiment *(How to test and characterize the design/prototype?)*	The system was evaluated through a case study addressing three defined research questions, assessing whether the AI-chatbot-based recommendation system could effectively deliver useful educational information to students.
Simulation/Test Data *(What parameters are determined?)*	The study is a deployed case-study system rather than a benchmark dataset; usage/interaction data came from the College of Industrial Technology in Thailand and its target student population.
Result / Conclusion *(What was the final result?)*	The system successfully addressed all three research questions and demonstrated that the AI-chatbot-based recommendation system could be accessed anytime via mobile devices and social media, effectively supporting educational recommendation use cases.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	The authors note the study's scope was limited to a single college case study, and recommend testing across a broader population and educational settings in future work.
Terminology *(List the common basic words frequently used in this research field)*	AI chatbot, educational recommendation system, mobile application, case study.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	Compared to purely technical chatbot-technique papers, this case study is more focused on real-world deployment and usability, showing that AI chatbot recommendation systems can be practically integrated into everyday mobile and social media platforms.
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can consider a similar mobile- and social-media-integrated deployment approach to maximize accessibility if our project involves an AI-based recommendation or advisory system.

Aspects	**Paper # 4**
Title / Question *(What is problem statement?)*	Title of paper: Systematic Review on Chatbot Techniques and Applications. Problem statement: Chatbot technology has evolved significantly over time, and there is a need to review and organize recent core research technologies for researchers entering the field.
Objectives / Goal *(What is looking for?)*	Review and introduce five key underlying technologies used in chatbot systems, to help chatbot researchers understand the current state of the field and identify future research directions.
Methodology/Theory *(How to find the solution?)*	The authors conduct a literature review and categorize chatbot technologies into five groups: natural language processing (NLP), pattern matching, semantic web, data mining, and context-aware computing.
Software Tools *(What program/software is used for design, coding and simulation?)*	Literature review methodology; no direct code implementation (the paper is itself a review/survey of techniques used by other systems).
Test / Experiment *(How to test and characterize the design/prototype?)*	As a review paper, the 'test' involves surveying and categorizing existing chatbot literature across the five identified technology areas rather than running an experiment.
Simulation/Test Data *(What parameters are determined?)*	The dataset consists of previously published chatbot research papers and systems that the authors reviewed and categorized (not a numerical or image dataset).
Result / Conclusion *(What was the final result?)*	The paper presents a clear taxonomy of five chatbot technology categories (NLP, pattern matching, semantic web, data mining, context-aware computing) along with a discussion of the latest advancement directions in each area.
Obstacles/Challenges *(List the methodological obstacles if authors mentioned in the article)*	Keeping the review up to date with the rapidly evolving chatbot technology landscape was noted as an inherent challenge of this kind of survey work.
Terminology *(List the common basic words frequently used in this research field)*	Natural Language Processing (NLP), pattern matching, semantic web, data mining, context-aware computing.
Review Judgment *(Briefly compare the objectives and results of all the articles you reviewed)*	This review complements the more applied AI-chatbot papers (tourism and educational recommendation) by providing the underlying technique-level taxonomy that explains what technologies power those applied systems.

Aspects	**Paper # 4 (cont.)**
Review Outcome *(Make a decision how to use/refer the obtained knowledge to prepare a separate and new methodology for your own research project)*	We can use this taxonomy to justify and document the specific NLP or chatbot techniques we choose when designing the AI component of our own system.